

Evaluation of the Entomopathogenic Fungi *Beauveria bassiana*, *Metarhizium anisopliae*, and *Isaria fumosorosea* for Control of *Sitophilus oryzae*

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ABSTRACT

The entomopathogenic fungi *Beauveria bassiana*, *Metarhizium anisopliae*, and *Isaria fumosorosea* were tested against the stored-grain pest *Sitophilus oryzae*. The fungi were isolated from the soil (from three locations in Attica, Greece: *B. bassiana* from Tatoion, *M. anisopliae* from Marathon, and *I. fumosorosea* from Aghios Stefanos) using larvae of *Galleria mellonella* as bait. Suspensions of 2.11×10^7 and 2.11×10^8 , 1.77×10^7 and 1.77×10^8 , and 1.81×10^7 and 1.81×10^8 conidia per ml of *B. bassiana*, *M. anisopliae*, and *I. fumosorosea*, respectively, were applied by three treatments: (i) sprayed on food and set in petri dishes with adults of *S. oryzae*, (ii) sprayed on adults of *S. oryzae* and set in petri dishes without food, and (iii) sprayed on adults of *S. oryzae* and set in petri dishes with food. The observed mortality of *S. oryzae* adults during the overall exposure period for the lowest, as well as for the highest, concentrations of *B. bassiana*, *M. anisopliae*, and *I. fumosorosea* ranged from 0 to 100%. Concentration was, in most of the cases tested, a critical parameter that determined the “speed of kill” of the exposed insect species for *B. bassiana* and *M. anisopliae*. Conversely, concentration was not that critical for *I. fumosorosea*, and survival was high in some of the combinations tested, even after 14 days of exposure. Both in the highest and the lowest concentrations of fungi, the mortality of *S. oryzae* adults was higher when the fungi were applied on adults than when they were applied on food. Higher mortality was observed when food was absent than when food was present, in most of the cases tested. The high efficacy levels recorded in the current study indicate that the tested fungi could be effective biocontrol agents against *S. oryzae*.

Alternatives to synthetic insecticides have been sought in response to the development of insecticide resistance and concerns about consumer health and environmental safety; biological control is one of the methods that have been developed. Agents for biological control of stored-product insect pests include hymenopterous parasitoids, predaceous insects or mites, and insect pathogens, such as entomopathogenic fungi, bacteria, viruses, protozoa, and nematodes (13, 24). The use of natural enemies and insect pathogens is considered compatible when biological control strategies in stored facilities are followed. Yet insect pathogens can be stored for a longer time than can the beneficial insects, and they infest a wide range of insect pest species in contrast to beneficial insects, which are host-specific. Generally, the presence of insects in cereals, even if they are there as biological control agents, is considered contamination, making the products unacceptable for consumption (13, 24). Among insect pathogens, entomopathogenic fungi are promising alternatives to chemical insecticides (17, 24); they have low mammalian toxicity, they exist naturally and

are environmentally safe, and they infect the insects by contact. Entomopathogenic fungi penetrate the cuticle, the insect dies, and then the fungi are recycled in the cadavers, thus returning more inoculum to the stored-product system (10, 24, 29, 30). Previous studies have examined the potential of entomopathogenic fungi against stored-product insects, with promising results. Searle and Doberski (27) reported complete mortality of *Oryzaephilus surinamensis* (L.) (Coleoptera: Silvanidae) adults exposed to *Beauveria bassiana* (Balsamo) Vuillemin (Ascomycota: Hypocreales) conidiospores at concentrations from 10^4 to 10^6 spores per ml after 26 days of exposure in wheat. Rice and Cogburn (25) investigated *B. bassiana* against adults of *Sitophilus oryzae* (L.) (Coleoptera: Curculionidae), *Rhyzopertha dominica* (F.) (Coleoptera: Bostrychidae), and *Tribolium castaneum* (Herbst) (Coleoptera: Tenebrionidae) on three types of substrates and recorded 80 to 100% mortality at the highest concentrations. Ekesi et al. (12) found that *B. bassiana*, *Metarhizium anisopliae* (Metschnikoff) Sorokin (Ascomycota: Hypocreales), and *Nomuraea rileyi* (Farlow) Sampson (Ascomycota: Hypocreales) caused 48 to 96%, 89 to 100%, and 33 to 95% mortality, respectively, of *Sitotroga cerealella* (Olivier) (Lepidoptera: Gelechiidae)

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TABLE 1. Repeated measures MANOVA parameters for main effects and associated interactions for mortality levels of *S. oryzae* adults^a

Fungus species	Source	df	F	P
Between exposure intervals				
<i>B. bassiana</i>	Treatment	1	323.0	<0.01
	Concentration	2	36.7	<0.01
	Treatment × concentration	2	19.3	<0.01
<i>M. anisopliae</i>	Treatment	1	398.5	<0.01
	Concentration	2	210.5	<0.01
	Treatment × concentration	2	13.6	<0.01
<i>I. fumosorosea</i>	Treatment	1	180.8	<0.01
	Concentration	2	0.3	0.60
	Treatment × concentration	2	0.0	0.99
Within exposure intervals				
<i>B. bassiana</i>	Exposure × treatment	10	102.2	<0.01
	Exposure × concentration	5	30.2	<0.01
	Exposure × treatment × concentration	10	22.6	<0.01
<i>M. anisopliae</i>	Exposure × treatment	10	122.7	<0.01
	Exposure × concentration	5	90.2	<0.01
	Exposure × treatment × concentration	10	101.4	<0.01
<i>I. fumosorosea</i>	Exposure × treatment	10	84.4	<0.01
	Exposure × concentration	5	3.3	<0.01
	Exposure × treatment × concentration	10	3.7	<0.01

^a MANOVA, multivariate analysis of variance. Error df = 114 for all fungi.

was determined by prodding with a brush to detect movement under an Olympus stereomicroscope (SZX9, Bacacos S.A., Athens, Greece) after 1, 2, 6, 7, 10, and 14 days of exposure. The brush was carefully washed after the examination of each dish. The entire procedure was repeated five times by preparing new wheat lots each time. All dead adults were removed from the petri dishes, put into new sterile petri dishes lined with wet filter paper, kept at 25°C, and inspected daily for symptoms of mycosis (8). Mycelium and conidia from all infected *S. oryzae* adults were examined under a stereomicroscope to confirm that they belonged to the fungal species with which they had been treated.

In the second treatment, 1,000 *S. oryzae* adults per plastic box (each 12 by 6 by 2 cm) were held at 4°C for 30 min to immobilize them (15) and then were sprayed with 1 ml of a suspension (one for each concentration of each fungus). When the application of each concentration for a certain fungus was completed, the same procedure was followed with the next fungus. Spraying was carried out with the same device mentioned previously. An additional box with 1,000 *S. oryzae* adults was left untreated (control). The internal walls of the boxes was covered by Fluon, to prevent insects from escaping. Four samples (10 *S. oryzae* adults each) were taken from each box (treated or control adults) and placed into petri dishes with a different brush. The internal walls of the boxes was covered by Fluon, to prevent insects from escaping. Following this, petri dishes were placed in incubators set at 27°C and 65% relative humidity during the entire experimental period. Adult mortality and mycosis were determined as mentioned previously. The entire procedure was repeated five times, with new groups of insects prepared each time.

In the third treatment, the procedure was similar to the second, except that insects were put in petri dishes that each contained 10 g of untreated wheat. The entire procedure was repeated five times, with new groups of insects prepared each time, as noted above.

Data analysis. In all cases, control mortality was very low (<5%) and, therefore, no correction was considered necessary for the mortality data. Data were analyzed separately for each of the

tested fungi according to repeated measures analysis and Wilks' lambda estimate. The repeated factor was exposure interval, and mortality was the response variable. Treatment and concentration were the main effects. The associated interactions of the main effects were incorporated in the analysis. Statistical analyses were done for each of the assessment periods using JMP 9 software (26). Means were separated by the Tukey-Kramer honestly significant difference test at $P = 0.05$ (28).

RESULTS

***B. bassiana*.** Mortality of *S. oryzae* adults was significantly affected by the exposure interval ($F = 62705$, $df = 5, 114$, $P < 0.01$). Repeated measures variables, for main effects and associate interactions, are presented in Table 1. After 1 and 2 days of exposure, mortality of *S. oryzae* adults was low for all treatments and concentrations and did not exceed 10% (Table 2). At the 6-day exposure interval, mortality significantly increased when fungus was applied on adults (with or without food) compared with when it was applied on food. Seven days later, all adults died when the fungus was applied on adults in the absence of food, at both concentrations. Still, the mortality of *S. oryzae* adults was low and did not exceed 7.5% in the treatment in which the fungus was applied on food. After 10 days of exposure, adult mortality was further increased when the fungus was applied on food, but it was significantly lower in the other treatments. After 14 days of exposure, all adults were dead at both concentrations studied.

***M. anisopliae*.** Mortality of *S. oryzae* adults was significantly affected by the exposure interval ($F = 28303$, $df = 5, 114$, $P < 0.01$). Repeated measures variables, for main effects and associate interactions, are presented in Table 1. As in the case of *B. bassiana*, after 1 and 2 days of exposure, mortality of *S. oryzae* adults was

TABLE 2. Mortality of *Sitophilus oryzae* adults exposed to *Beauveria bassiana* in two concentrations by three treatments^a

Exposure (days)	Concn (conidia per ml)	Sprayed on food	Sprayed on adults		F	P
			No food	Food		
1	2.11×10^7	0.0 ± 0.0 b	0.0 ± 0.0 b	5.0 ± 2.0 A a	6.3	<0.01
1	2.11×10^8	0.0 ± 0.0 b	0.0 ± 0.0 b	2.5 ± 1.0 A a	6.3	<0.01
F		—	—	1.3		
P		—	—	0.27		
2	2.11×10^7	0.0 ± 0.0 b	2.5 ± 1.0 A ab	10.0 ± 4.0 A a	4.8	0.01
2	2.11×10^8	0.0 ± 0.0 b	0.0 ± 0.0 B	10.0 ± 2.8 A a	12.7	<0.01
F		—	6.3	0.0		
P		—	0.02	1.00		
6	2.11×10^7	2.5 ± 1.0 A c	67.5 ± 1.9 A a	30.0 ± 4.3 B b	138.7	<0.01
6	2.11×10^8	2.5 ± 1.0 A b	77.5 ± 8.9 A a	82.5 ± 3.4 A a	65.2	<0.01
F		0.0	1.2	92.1		
P		1.00	0.28	<0.01		
7	2.11×10^7	5.0 ± 2.0 A c	100.0 ± 0.0 a	52.5 ± 5.5 B b	191.3	<0.01
7	2.11×10^8	7.5 ± 1.9 A b	100.0 ± 0.0 a	100.0 ± 0.0 A a	2,286.3	<0.01
F		0.8	—	75.4		
P		0.37	—	<0.01		
10	2.11×10^7	40.0 ± 8.1 A b	100.0 ± 0.0 a	87.5 ± 5.0 B a	33.2	<0.01
10	2.11×10^8	60.0 ± 6.5 A b	100.0 ± 0.0 a	100.0 ± 0.0 A a	38.0	<0.01
F		3.7	—	6.3		
P		0.06	—	0.02		
14	2.11×10^7	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	—	—
14	2.11×10^8	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	—	—
F		—	—	—		
P		—	—	—		

^a Treatments: sprayed on food and set in petri dishes with adults, sprayed on adults and set in petri dishes without food, or sprayed on adults and set in petri dishes with food, for 1, 2, 6, 7, 10, and 14 days. Values are mean percentages $\pm$ standard errors. Within each column, means followed by the same uppercase letter are not significantly different; in all cases, $df = 1, 39$ (Tukey-Kramer HSD test; $P = 0.05$). Within each row, means followed by the same lowercase letter are not significantly different; in all cases, $df = 2, 59$ (Tukey-Kramer HSD test; $P = 0.05$). —, no analysis was performed.

low for all treatments and concentrations and did not exceed 5% (Table 3). At the 6-day exposure interval, all *S. oryzae* adults were dead in the case of absence of food at the highest concentration, whereas the mortality in the other treatments remained low and did not exceed 32.5%. After 7 days of exposure, mortality was still significantly lower when the fungi were applied on food in comparison to the other treatments. Ten days later, mortality of *S. oryzae* adults did not exceed 25% when the fungus was applied on food at the lowest concentration. After 14 days of exposure, all adults were dead when the fungus was applied on adults, with or without food, at both concentrations studied.

***I. fumosorosea*.** Mortality of *S. oryzae* adults was significantly affected by the exposure interval ($F = 915$, $df = 5, 114$, $P < 0.01$). Repeated measures variables, for main effects and associate interactions, are presented in Table 1. After 1 day of exposure, no mortality of *S. oryzae* adults was recorded; whereas, after 2 and 6 days of exposure, mortality was low for all treatments and concentrations and did not exceed 5 and 17.5%, respectively (Table 4). Seven and 10 days later, all adults died in the absence of food at both concentrations tested. Even in this case, significantly fewer *S. oryzae* adults died than in the other two treatments. At 14 days of exposure, adult mortality was significantly

lower when fungus was applied on food than when it was applied on adults without food.

DISCUSSION

Since 2000, there have been numerous studies on the use of entomopathogenic fungi in stored-product protection (5, 6, 14, 17, 19, 21, 24). In most of these studies, the fungi were applied directly on the commodity, particularly grain, a method that is likely to affect the efficacy of the application. For instance, Michalaki et al. (21), by using *M. anisopliae* in stored grains against *Tribolium confusum* Jacquelin du Val (Coleoptera: Tenebrionidae) larvae, found that the “speed of kill” was relatively low and survival was high at 7 days of exposure. Similarly, Kavallieratos et al. (17), by using the same fungus against adults of *R. dominica*, *S. oryzae*, and *T. confusum*, noted considerable survival even after 14 days of exposure, regardless of the concentration of fungus applied. In our study, concentration was, in most of the cases tested, a critical parameter that determined the “speed of kill” of the exposed insect species, at least for *B. bassiana* and *M. anisopliae*. Conversely, concentration was not a critical factor for *I. fumosorosea*, and survival was high in some of the combinations tested, even after 14 days of exposure. Despite the fact that there is no study available that compares the three fungal species against *S. oryzae*,

TABLE 3. Mean mortality of *Sitophilus oryzae* adults exposed to *Metarhizium anisopliae* in two concentrations by three treatments^a

Exposure (days)	Concn (conidia per ml)	Sprayed on adults			F	P
		Sprayed on food	No food	Food		
1	1.77×10^7	0.0 ± 0.0	0.0 ± 0.0 B	0.0 ± 0.0	—	—
1	1.77×10^8	0.0 ± 0.0 b	2.5 ± 1.0 A a	0.0 ± 0.0 b	6.3	<0.01
F		—	6.3	—		
P		—	0.02	—		
2	1.77×10^7	2.5 ± 1.0 A a	0.0 ± 0.0 B b	0.0 ± 0.0 B b	6.3	0.01
2	1.77×10^8	0.0 ± 0.0 B b	5.0 ± 1.2 A a	2.5 ± 1.0 A ab	8.1	<0.01
F		6.3	19.0	6.3		
P		0.02	<0.01	0.02		
6	1.77×10^7	5.0 ± 2.0 A a	12.5 ± 3.8 B a	10.0 ± 4.0 B a	1.3	0.28
6	1.77×10^8	2.5 ± 1.0 A c	100.0 ± 0.0 A a	32.5 ± 1.0 A b	3,790.5	<0.01
F		1.3	541.3	30.2		
P		0.27	<0.01	<0.01		
7	1.77×10^7	5.0 ± 1.2 A c	90.0 ± 2.8 B a	32.5 ± 1.0 B b	302.2	<0.01
7	1.77×10^8	7.5 ± 3.0 A c	100.0 ± 0.0 A a	77.5 ± 1.0 A b	1,498.5	<0.01
F		0.5	12.7	1,026.0		
P		0.49	<0.01	<0.01		
10	1.77×10^7	25.0 ± 6.2 B b	92.5 ± 3.0 B a	85.0 ± 3.8 B a	66.8	<0.01
10	1.77×10^8	80.0 ± 8.0 A b	100.0 ± 0.0 A a	100.0 ± 0.0 A a	6.3	<0.01
F		30.0	3.3	15.6		
P		<0.01	0.02	<0.01		
14	1.77×10^7	95.0 ± 0.0 B b	100.0 ± 0.0	100.0 ± 0.0	6.3	<0.01
14	1.77×10^8	100.0 ± 0.0 A	100.0 ± 0.0 a	100.0 ± 0.0 a	—	—
F		6.3	—	—		
P		0.02	—	—		

^a Treatments: sprayed on food and set in petri dishes with adults, sprayed on adults and set in petri dishes without food, or sprayed on adults and set in petri dishes with food, for 1, 2, 6, 7, 10, and 14 days. Values are mean percentages ± standard errors. Within each column, means followed by the same uppercase letter are not significantly different; in all cases, df = 1, 39 (Tukey-Kramer HSD test; $P = 0.05$). Within each row, means followed by the same lowercase letter are not significantly different; in all cases, df = 2, 59 (Tukey-Kramer HSD test; $P = 0.05$). —, no analysis was performed.

previous reports suggested that *B. bassiana* and *M. anisopliae* may be more effective than *I. fumosorosea* against some major stored-product insect species. For example, Michalaki et al. (22) found that *I. fumosorosea* was not effective against adults and larvae of *T. confusum* and larvae of *Ephestia kuehniella* Zeller (Lepidoptera: Pyralidae). On the other hand, Lord (19) reported that diatomaceous earth synergized *B. bassiana* against adults of *R. dominica* and *O. surinamensis* in wheat. Similarly, Vassilakos et al. (31) showed that inclusion of diatomaceous earth had an additive effect on *B. bassiana* efficacy in certain combination treatments against *R. dominica* in wheat. Kavallieratos et al. (17) recorded 82 and 94% mortality of *S. oryzae* and *R. dominica* adults in wheat treated with *M. anisopliae* at the concentration of 8×10^8 conidia per kg after 14 days of exposure.

The method of application of the fungal formulation was the most important factor in its efficacy against *S. oryzae* adults. Mortality was faster when the fungus was applied directly on the adults. The “speed of kill” is a crucial parameter in the use of entomopathogenic fungi for stored-grain protection, since fungi are generally slow acting, and oviposition with the concomitant progeny production is likely to occur before death (24). Previous studies provide interesting information on this issue. For

example, Cherry et al. (9) reported 100% mortality of *Callosobruchus maculatus* (F.) (Coleoptera: Bruchidae) adults dipped for 5 s into conidial suspension of *B. bassiana* or *M. anisopliae* after 6 and 8 days of exposure. Furthermore, Kassa et al. (16) found that *S. zeamais* adults dipped in certain isolates of *B. bassiana* suspension for 5 s had 100% mortality after 4 days of exposure. Similar results were also obtained for *P. truncatus* treated with certain *B. bassiana* and *M. anisopliae* isolates. Of the application methods examined here, direct application of the fungus on insect bodies is more effective than contact of the insects with the fungus-treated substrate. This difference is more evident at exposures ≥ 6 days, whereas at shorter exposures, all treatments had the same efficacy level. Hence, based on the present findings, it is postulated that at longer exposures, e.g., >14 days, the application on wheat could result in 100% mortality, at least in the cases of *M. anisopliae* and *I. fumosorosea*.

The presence of food when fungi were applied directly on the insects led to decreased mortality, at least in most of the cases tested, perhaps because food enabled the exposed individuals to moderate stress and partially recover. Studies of various types of insecticides have illustrated the importance of food in the efficacy of these compounds against stored-product insects. For example, White and

TABLE 4. Mean mortality of *Sitophilus oryzae* adults exposed to *Isaria fumosorosea* in two concentrations by three treatments^a

Exposure (days)	Concn (conidia per ml)	Sprayed on food	Sprayed on adults		F	P
			No food	Food		
1	1.81×10^7	0.0 ± 0.0	0.0 ± 0.0 B	0.0 ± 0.0	—	—
1	1.81×10^8	0.0 ± 0.0	0.0 ± 0.0 B	0.0 ± 0.0	—	—
F		—	—	—		
P		—	—	—		
2	1.81×10^7	2.5 ± 1.0 A ab	5.0 ± 1.2 A a	0.0 ± 0.0 B b	8.1	<0.01
2	1.81×10^8	0.0 ± 0.0 B b	5.0 ± 1.2 A a	2.5 ± 1.0 A ab	8.1	<0.01
F		6.3	0.0	6.3		
P		0.02	1.0	0.02		
6	1.81×10^7	2.5 ± 1.0 A b	12.5 ± 3.0 A a	10.0 ± 4.0 A a	4.2	0.02
6	1.81×10^8	5.0 ± 1.2 A b	17.5 ± 3.4 A a	12.5 ± 2.5 A a	6.2	<0.01
F		2.7	1.2	0.3		
P		0.11	0.28	0.60		
7	1.81×10^7	5.0 ± 2.0 A c	100.0 ± 0.0 a	22.5 ± 4.1 B b	370.1	<0.01
7	1.81×10^8	10.0 ± 2.3 A c	100.0 ± 0.0 a	37.5 ± 5.7 A b	168.7	<0.01
F		2.7	—	4.6		
P		0.11	—	0.04		
10	1.81×10^7	27.5 ± 7.1 A c	100.0 ± 0.0 a	52.5 ± 2.5 B b	71.1	<0.01
10	1.81×10^8	37.5 ± 8.9 A c	100.0 ± 0.0 a	67.5 ± 6.8 A b	23.3	<0.01
F		0.8	—	4.2		
P		0.39	—	0.04		
14	1.81×10^7	82.5 ± 7.0 A b	100.0 ± 0.0 a	90.0 ± 4.0 B ab	3.6	0.03
14	1.81×10^8	85.5 ± 4.8 A b	100.0 ± 0.0 a	100.0 ± 0.0 A a	9.0	<0.01
F		0.1	—	6.3		
P		0.73	—	0.02		

^a Treatments: sprayed on food and set in petri dishes with adults, sprayed on adults and set in petri dishes without food, or sprayed on adults and set in petri dishes with food, for 1, 2, 6, 7, 10, and 14 days. Values are mean percentages ± standard errors. Within each column, means followed by the same uppercase letter are not significantly different; in all cases, df = 1, 39 (Tukey-Kramer HSD test; $P = 0.05$). Within each row, means followed by the same lowercase letter are not significantly different; in all cases, df = 2, 59 (Tukey-Kramer HSD test; $P = 0.05$). —, no analysis was performed.

Loschiavo (33) reported that silica aerogel became less efficacious against *Oryzaephilus mercator* (Fauvel) (Coleoptera: Silvanidae) when food (rolled oats, ground rolled oats, or ground white bread crumbs) was available. Arthur (2) reported that the availability of food (flour or pine sawdust) increased the survival of *T. castaneum* adults in concrete treated with cyfluthrin. Arthur (3) found that the presence of flour reduced the mortality of adults of *T. castaneum* and *T. confusum* that were exposed to diatomaceous earth. Hence, in a storage facility, the presence of areas with food, such as small food sources in cracks and crevices, may decrease the efficacy of a given insecticidal application. This is likely to occur even if this food is treated with insecticide (4). Our work clearly indicates that availability of food may increase survival of the exposed *S. oryzae* to entomopathogenic fungi. This effect is likely to be the same, even in those cases in which food contains entomopathogenic fungi. Consequently, in a given storage facility, sanitation that leads to the drastic reduction of food refugia may be more crucial than the application of insecticide. This is particularly important in the case of entomopathogenic fungi, since these compounds are slow acting and food sources may notably prolong survival.

The results indicate that the entomopathogenic fungi tested here were effective against *S. oryzae*. Generally, *B.*

bassiana and *M. anisopliae* were more effective than *I. fumosorosea*. Moreover, the application of the fungi directly on insects is more effective than their application on grain, which suggests that these compounds could be better used in an “attract and kill” baiting station approach than by treating the entire grain mass. Such an application would also be more consumer friendly than the “pathogen-treated food” approach. The simultaneous use of other reduced-risk compounds that would increase the “speed of kill” of fungi should be further evaluated.

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